

## src/plotting.py

88 lines

```
1 """
2 plotting.py
3 =====
4 Shared "house style" for every figure in the project, so step-test plots,
5 model-validation plots and scenario plots all look like they belong to one
6 polished submission.
7 """
8
9 from __future__ import annotations
10
11 import matplotlib
12
13 matplotlib.use("Agg") # headless, no display needed for report generation
14 import matplotlib.pyplot as plt
15 import numpy as np
16
17 # -----
18 # Palette
19 # -----
20 COLOR_CHOKE = "#6b7280" # slate grey
21 COLOR_Q = "#2563eb" # blue
22 COLOR_TARGET = "#111827" # near-black, dashed
23 COLOR_WHP = "#059669" # green
24 COLOR_FLP = "#d97706" # amber
25 COLOR_BHP = "#dc2626" # red
26 COLOR_SAFE = "#22c55e"
27 COLOR_LIMIT = "#dc2626"
28 COLOR_BAND = "#22c55e"
29
30 plt.rcParams.update(
31 {
32     "figure.facecolor": "white",
33     "axes.facecolor": "white",
34     "axes.edgecolor": "#374151",
35     "axes.labelcolor": "#111827",
36     "axes.titlesize": 12,
37     "axes.titleweight": "bold",
38     "axes.grid": True,
39     "grid.color": "#e5e7eb",
40     "grid.linewidth": 0.8,
41     "font.size": 10,
42     "font.family": "DejaVu Sans",
43     "legend.frameon": False,
44     "xtick.color": "#374151",
45     "ytick.color": "#374151",
46     "figure.dpi": 130,
47     "savefig.dpi": 150,
48     "savefig.bbox": "tight",
49 }
50 )
51
52
53 def style_axis(ax, title=None, ylabel=None, xlabel=None):
54     if title:
55         ax.set_title(title, loc="left")
56
57     if ylabel:
58         ax.set_ylabel(ylabel)
59
60     if xlabel:
61         ax.set_xlabel(xlabel)
62
63     ax.spines["top"].set_visible(False)
64     ax.spines["right"].set_visible(False)
65     return ax
66
67
68 def shade_safe_band(ax, lo, hi, margin_lo=None, margin_hi=None, label="Safe band"):
69     """Green safe band between hard limits, with limit lines drawn on top."""
70     ax.axhspan(lo, hi, color=COLOR_BAND, alpha=0.08, zorder=0, label=label)
```

```

68 ax.axhline(lo, color=COLOR_LIMIT, lw=1.3, ls="--", zorder=1)
69 ax.axhline(hi, color=COLOR_LIMIT, lw=1.3, ls="--", zorder=1)
70 if margin_lo is not None:
71 ax.axhline(lo + margin_lo, color="#f59e0b", lw=0.9, ls=":", zorder=1)
72 if margin_hi is not None:
73 ax.axhline(hi - margin_hi, color="#f59e0b", lw=0.9, ls=":", zorder=1)
74
75
76 def ramp_envelope(ax, t, u0, max_move, ts):
77 """Draw the feasible ramp-rate cone from the initial choke position."""
78 steps = (np.asarray(t) - t[0]) / ts
79 hi = np.clip(u0 + max_move * steps, 0, 100)
80 lo = np.clip(u0 - max_move * steps, 0, 100)
81 ax.fill_between(t, lo, hi, color=COLOR_CHOKE, alpha=0.08, zorder=0,
82 label="Ramp envelope")
83
84
85 def savefig(fig, path):
86 fig.savefig(path)
87 plt.close(fig)
88

```